arXiv:2608.13560v1 [cs.CV] 13 Aug 2026

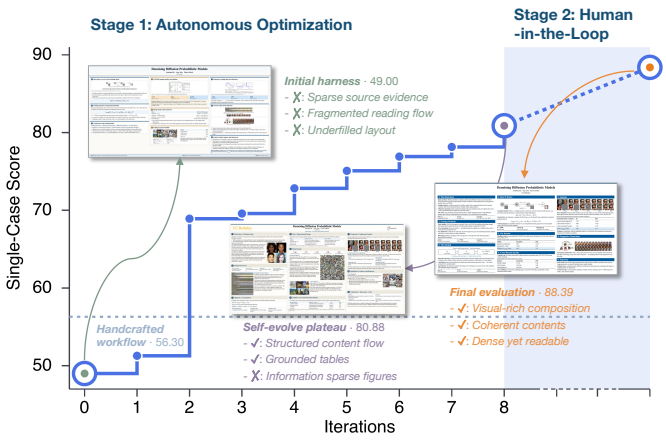
AutoDesign: Meta-Harness Optimization for Long-Horizon Agentic Design

Yaxin Luo[1,2*], **Haobin Jiang**[1*], **Jialv Zou**[1,3], **Xu Huang**[1,4], **Wenhao Yan**[1,5], **Haodong Li**[1,6], **Zhengrong Yue**[1,7], **Jing Li**[1], **Xiaofu Chen**[2], **Xiaohan Zhao**[2], **Jiacheng Liu**[2], **Jiacheng Cui**[2], **Zhiqiang Shen**[2✉], **Xiaotong Li**[1✉,†]

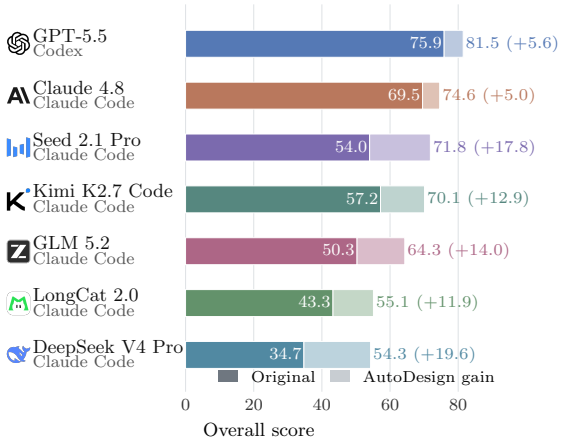
[1]Meituan, [2]MBZUAI, [3]Huazhong University of Science and Technology, [4]Peking University, [5]Tsinghua University, [6]The Chinese University of Hong Kong, [7]Shanghai Jiao Tong University
[*]Equal contribution, [†]Project lead

Transforming multimodal sources into condensed and structured media outputs can be fundamentally conceptualized as a long-horizon agentic process centered on a model-harness system. While an ideal harness system should align with human design priors and accumulate reusable experience through empirical exploration to drive recursive self-improvement, existing paradigms remain static and fall short of this capability. In this paper, we present AutoDesign, a framework that aligns with human design priors, where a meta-harness optimizer guides a code agent to recursively improve harness based on rollout feedback. To instantiate and evaluate this framework, we focus on the academic paper-to-poster generation task and introduce PosterBench, comprising a 100-paper Main Track spanning five disciplines and PosterBench-mini, a shared 10-paper subset for controlled evaluation. On the PosterBench Main Track, AutoDesign achieves the highest score of 78.32, surpassing the closed-source commercial system Claude Design by 7.45 points. Across seven controlled code agent–model configurations, integrating the learned DesignHarness consistently improves performance, increasing the average PosterBench Score from 54.99 to 67.39 (+12.4%). In a fully autonomous long-horizon loop, it executes 253 tool calls and 11 editing turns within 40 minutes for under $3, reaching average conference-poster quality in human evaluation. A system-blind human study further demonstrates that AutoDesign achieves the highest human preference among evaluated systems.

Project Website https://autodesign.designanything.ai/
Code Repository https://github.com/Yaxin9Luo/AutoDesign
Demo Page https://designanything.ai/



(a) Meta-harness optimization trace



(b) Performance gains from DesignHarness

Figure 1 AutoDesign progressively improves the design harness and the quality of the artifacts. (a) Score of the poster generated by the design harness for one representative paper, tracked across meta-harness iterations. Autonomous optimization improves the initial harness before reaching a plateau, after which human guidance redirects the search and yields a further gain. (b) The optimized harness, DesignHarness, improves all Coding Agents on PosterBench by 5.0 to 19.6 points, achieving a best overall score of 81.5.

Correspondence: ✉lixiaotong.edu@gmail.com ✉zhiqiang.shen@mbzuai.ac.ae
Work done during Yaxin's internship at Meituan.